Two contradicting literatures, one scalar

When Does Shift-Invariance Permit Adversarial Robustness?

A threat-matched margin-to-Lipschitz diagnostic η/L

Anass Al Ammiri

Technical University of Munich

Thesis: the "invariance hurts robustness" and "invariance helps it" claims are two regimes of one gauge-free, threat-matched ratio η/L; the field's invariance metric does not predict robustness, this ratio does, and the "helps" claims are weak-attack artifacts.

The puzzle: same word, opposite verdicts

Does making a CNN more shift-invariant help or hurt adversarial robustness?

"Invariance HURTS"

Ge et al. 2021. A fully shift-invariant classifier can be far *less* robust than a plain one. On a one-pixel problem the achievable margin collapses from a constant to $\mathcal{O}(1/\sqrt{d})$.

Kamath 2021: a provable spatial-vs-adversarial trade-off.

"Invariance HELPS"

Anti-aliasing / TIPS / Wang 2025. Reducing aliasing, learned translation-invariant pooling, and rotation/scale equivariance are *reported to improve* adversarial robustness, argued through an orbit-invariant Lipschitz bound.

"A contradiction in the literature is a flashing sign that some shared assumption is wrong, and the assumption is almost always a word doing too much work. Here the overloaded word is *invariance*."

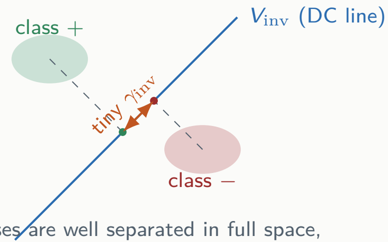
Takeaway. These are not all wrong. They measure different regimes of the *same* underlying quantity. We put them on one axis.

Why invariance can quietly destroy the signal

The linear case is brutally clean

A shift-invariant *linear* classifier must answer the same for a signal and every shift of it. That forces its weights onto the shift group's fixed subspace:

- "The **only** linear shift-invariant feature is the **mean**." For images: a linear invariant sees nothing but the *average colour*.
- On Ge's "black vs. white pixel", it keeps only the DC component; the margin collapses from a constant to $2/\sqrt{d}$ ($\sim 28\times$ at $d = 784$).
- **Position lives entirely in the phase**: the power spectrum (magnitude only) is *blind* to location, so even it cannot separate $\pm e_j$.



classes are well separated in full space, but nearly overlap once projected

Takeaway. Invariance keeps only the signal that survives projection onto the invariant subspace; if little survives, robustness must trade off.

The one idea: a margin-to-Lipschitz ratio η/L

Margin that survives the invariance, over its sensitivity to the attack

$$r_2 \;\geq\; \frac{\eta}{L} \;=\; \frac{\overbrace{\eta}^{\text{margin that survives invariance}}}{\underbrace{L}_{\text{its sensitivity to the attack}}}$$

The certificate. A model is certifiably safe over a ball when it is *decisive* (big margin η) *and gentle* (small Lipschitz L). Robustness attaches to the *ratio*, not to the margin or Lipschitz term alone.

One axis for both camps.

- Large η/L: invariance and robustness *coexist*.
- Small surviving margin: they *trade off*.
- Ge / Kamath = the small-η/L corner.

Two properties make η/L the right scalar

Gauge-free and threat-matched

Rescale the logits $f \mapsto c\,f$: both η and L scale by c, so η/L is unchanged.

The margin and the Lipschitz constant are each *gauge-dependent*, so "robustness improved because the margin grew" is only well-posed once a gauge is fixed. The *ratio* is the honest, gauge-free quantity.

An attack in norm ℓ_p is resisted by sensitivity in the *dual* norm ℓ_q.

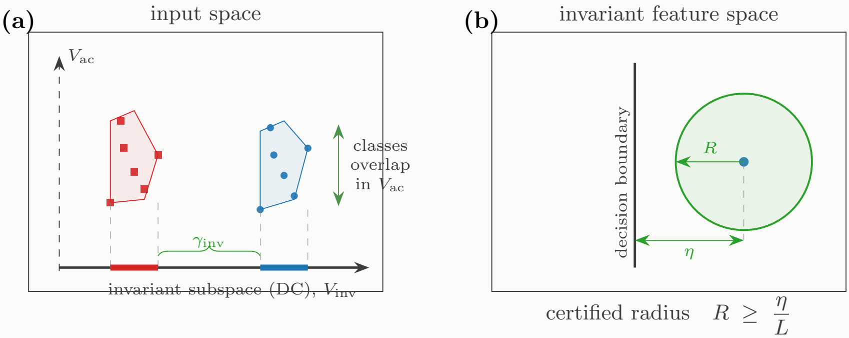
So we always match the Lipschitz term to the attack: $L_q = \mathbb{E}\|\nabla M\|_q$. For an ℓ_∞ attack, use the ℓ_1 gradient norm. Get this wrong and the ratio can even *anti*-predict.

Read it off a trained model on clean inputs: $\eta = \mathbb{E}[\text{logit margin}], \quad L_q = \mathbb{E}\|\nabla(\text{margin})\|_q.$

Takeaway. One organizing principle: invariance changes robustness *only* through η/L, and the attack's norm decides which Lipschitz constant counts.

Structural theory, in one picture

Invariant margin = separation of the data projected onto the invariant subspace



(a) An invariant classifier sees the data only through its projection onto $V_{\rm inv}$. What separation remains is the invariant margin $\gamma_{\rm inv}$; signal in the orthogonal (AC) subspace is invisible.

(b) A surviving margin η certifies $r_2 \geq \eta/L$. Invariance helps robustness *only* when the discriminative signal survives the projection with positive margin.

Three structural results behind η/L

The invariant linear max-margin *equals* the separation of the DC-projected data, $\gamma_{\text{inv}} = \frac{1}{2}\,\mathrm{dist}\big(\mathrm{conv}\,\Pi_{\text{inv}}X_+,\ \mathrm{conv}\,\Pi_{\text{inv}}X_-\big)$. Recovers Ge's $1/\sqrt{d}$ collapse as a special case; holds for any finite dataset and any finite symmetry group.

A convolution + quadratic activation + global pooling computes the *power spectrum* $\Phi_w(x) = \frac{1}{d}\sum_k |\hat{w}_k|^2|\hat{x}_k|^2$: a shift-invariant nonlinear feature that can separate data linear invariance cannot. Invariance can help *through the right nonlinear feature*.

Under a reachability condition the one-sided certificate completes to $\eta/L \le r_2 \le \eta/\alpha$, with $\kappa = L/\alpha$ (exact, κ=1, for linear invariants). So η/L determines the robust radius up to a condition number κ.

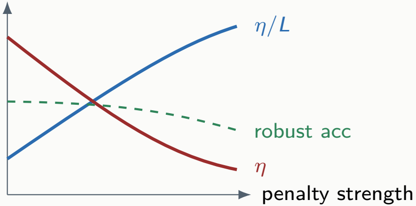
η/L is a diagnostic, *not* a trainable target

The scale-invariance that makes it honest also makes it untrainable

Because η/L is scale-invariant, maximizing it directly is ill-posed: the global optimizer is the constant classifier ($\nabla M \equiv 0$), which classifies at chance.

An exact identity (Euler) ties the margin gradient to the margin itself: $\|\nabla M\| \geq M/\|x\|$. Lowering the gradient drags the margin down with it.

Four first-order attempts to raise η/L (isotropic / anisotropic gradient penalties, ratio maximization, directional penalty) all collapse the margin and leave robustness flat or worse.

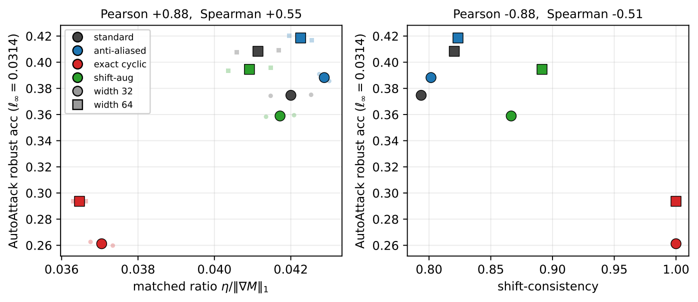


diagnostic moves,
robustness does not

Takeaway. You can *read* η/L off a model to rank it; you cannot *optimize* it into robustness.

Main empirical result: the dissociation

The field's invariance metric does not predict robustness; η/L does



Left. Threat-matched $\eta/\|\nabla M\|_1$ tracks AutoAttack robust accuracy on adversarially trained CIFAR-10 (Pearson $+0.88$).

Right. Shift-consistency anti-predicts it (-0.88). The exactly invariant model (red) has consistency 1.0 and the *lowest* robust accuracy.

Takeaway. Perfect shift-consistency, worst robustness. Consistency is the *wrong* axis; the threat-matched ratio is the right one.

The dissociation is not a one-off

Same two laws across datasets, threat models, and a ResNet backbone

condition	training / threat	consist. vs rob.	matched η/L vs rob.
CIFAR-10 dissection	standard, r_2	-0.33	$+0.998$
CIFAR-10 dissection	PGD-AT, ℓ_∞	-0.88	$+0.88$
CIFAR-10 dissection	PGD-AT, ℓ_2	-0.81	$+0.88$
MNIST dissection	PGD-AT, ℓ_∞	-0.44	$+0.92$
Fashion-MNIST dissection	PGD-AT, ℓ_∞	-0.92	$+0.71$
ResNet-18 (with TIPS)	PGD-AT, ℓ_∞	-0.79	$+0.91$
ImageNet-100	Fast-AT, ℓ_∞	-0.87	$+0.90$
diffusion *data* axis	PGD-AT, ℓ_∞	near-const	$+0.85$

Consistency: uninformative to anti-predictive. Threat-matched η/L: predicts throughout.
ImageNet-100: clean accuracy is *flat* across arms, so the deficit is margin geometry, not underfitting.

The strongest result: “invariance helps” is a weak-attack artifact

We retrained the contrary papers’ own models under their own recipe

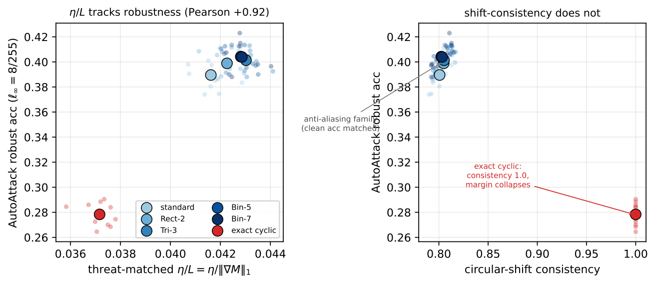
		weak attack		strong attack	
arm (standard-trained)	consist.	FGSM_1	FGSM_2	AA_2	AA_4
standard	0.965	0.477	0.244	0.006	0.000
anti-aliased	0.978	0.472	0.262	0.002	0.000
exact invariant (APS)	1.000	0.497	0.277	0.004	0.000
shift-augmented	0.973	0.532	0.315	0.004	0.000
TIPS + aux losses	0.973	0.493	0.300	0.004	0.000

- Under FGSM the “invariance helps” ordering *does* appear (invariant arms above the baseline).
- Under AutoAttack every arm collapses to ≤ 0.006 at $\varepsilon = 2/255$ — there is no robustness left to order.
- Same for Wang 2025’s rotation-equivariant net: FGSM 28.5% at their headline budget, but 0.1% under threat-matched AutoAttack.

$\Rightarrow$ **The “invariance helps” correlation lives only under weak attacks on undefended models; it vanishes when the attack converges.**

Honest scope: invariance *can* help, when it does not cost margin

A graded anti-aliasing control at matched clean accuracy



Left. Under adversarial training, graded blur-pooling raises robust accuracy at matched clean accuracy, and η/L orders all six arms ($+0.92$): invariance buys robustness *through the margin*.

Right. Consistency is non-monotone: flat across the anti-aliasing family, then jumps to 1.0 at the exact-cyclic arm, which is the *least* robust.

Takeaway. We do not claim invariance always hurts. η/L, not consistency, tells you *which* invariance helps.

What it means

1. **One axis unifies a split field.** "Invariance hurts" and "invariance helps" are the small- and large-η/L regimes of one gauge-free, threat-matched ratio.
2. **Stop measuring shift-consistency for robustness.** It does not order adversarial robustness and *anti*-predicts it under adversarial training. As an attack-free selection rule it costs $\sim$8.8 pts of robust accuracy vs. the oracle; picking by η/L costs 0.06.
3. **Distrust "invariance helps" claims measured under weak attacks.** The effect is an FGSM artifact; it vanishes under AutoAttack and under adversarial training.
4. **η/L is a diagnostic, not a defense.** Read it to rank operators; do not try to optimize it — that collapses the margin.

Honest scope: η/L is a lower-bound certificate (no universal converse), predicts *within* an intervention axis, and the headline is the dissociation, not the near-perfect η/L–radius fit (that part is partly definitional).

Does it generalize?

The same dissociation in frozen VLM encoders, LLMs, and transformers

Generalization 1: frozen vision-language encoders

η/L on clean images ranks eleven towers by AutoAttack robustness; consistency does not

predictor (clean, attack-free)	tower rank vs robust acc	after removing "is-AT"
threat-matched η/L	$+0.953$	$+0.51$
cosine "consistency"	$+0.81$	$+0.15$ (only detects AT)
shift-consistency (prediction)	$+0.37$ n.s.	–
clean accuracy	-0.07	–

Per image (the powered claim): η/L orders the certified radius at Spearman $+0.78$ pooled over robust towers, and $+0.54$–$+0.79$ within each ordinary CLIP tower — so it is *not* an artifact of adversarial training. Consistency: $+0.12$.

Selection rule. Pick the tower by $\eta/L \to 5$ pts of robust accuracy behind the oracle. Pick by consistency or clean accuracy $\to$ 88 pts behind (both choose the zero-robust tower).

Takeaway. The CIFAR dissociation is not a CIFAR quirk: it holds through *frozen* CLIP / RobustVLM / DINOv2 towers, attack-free.

Generalization 2: instruction-tuned language models

Sensitivity (η/L) and excessive invariance (ρ_G) are *independent* axes

- **Two robustness notions, one model.** η/L = how far a correct decision sits from the boundary vs how fast the margin moves. ρ_G = the size of the smallest *meaning-changing* edit (antonym, negation, harmful→benign) the model wrongly ignores — an orbit-flip radius, with a certificate $\rho_G \geq$ the oracle-robust radius.
- **They are orthogonal.** Across 1600 prompts, Spearman$(\eta/L, \rho_G) = -0.025$ $[-0.09, +0.04]$: sensitivity to worst-case perturbations and invariance to large meaning-changing edits are *unrelated* in LLMs. (This inverts the certified-NLP direction, which certifies staying unchanged as a virtue.)
- **Excessive invariance, decomposed honestly.** On sentiment, a genuine 54/46 classifier wrongly ignores a meaning flip on 0.67 of correct items (clean antonyms 0.673) — real excessive invariance. On negation-NLI and refusal it is a constant classifier ("entailment" on 100%, "refuse" on 97%; *Qwen is not*), a boundary condition we report as such, not a headline.

Takeaway. In text the two notions become measurably distinct axes — the dissociation restated with modality-native instruments.

What transfers, and where we stop honestly

- ✓ **The weak-attack artifact replicates in VLMs.** Every non-robust CLIP tower shows FGSM robust accuracy 0.16–0.46 that AutoAttack drives to 0.0 — the same "invariance helps is an FGSM mirage" we saw on CIFAR, now in foundation encoders.
- ✓ **One gauge-free ratio, three settings.** CIFAR classifiers, frozen VLM towers, and LLMs: η/L predicts, invariance does not.
- ? **Where it stops.** A *first-token* refuse margin does *not* predict an LLM's *multi-token* jailbreak radius (Spearman -0.17). We trace this to a threat-model mismatch and test the matched continuation margin — the honest boundary of the encoder-level result, not a contradiction of it.
- $\Rightarrow$ **One cross-modal paper.** Vision + VLM primary, LLM as the generalization frontier. Target: a main-track venue; deeper representation analysis $\rightarrow$ ICLR.

Takeaway. The claim is a property of foundation models, not a dataset artifact — stated with its limits.

How to pitch this

Everything a student needs to give the talk cold

Pitch kit (1/3): the thesis and the 3-beat story

The one-sentence thesis

Two literatures fight over whether shift-invariance helps or hurts adversarial robustness; they are two regimes of one *gauge-free, threat-matched* margin-to-Lipschitz ratio η/L, the field's invariance metric does not predict robustness while η/L does, and the "invariance helps" claims are weak-attack artifacts.

The 3-beat narrative:

Beat 1. **Tension.** Same word, opposite verdicts. Ge: invariance hurts. Anti-aliasing / TIPS / Wang: invariance helps. Nobody put them on one axis.

Beat 2. **Resolution.** One scalar η/L = margin that survives the invariance, over its sensitivity in the attack's dual norm. Large ratio $\Rightarrow$ they coexist; small surviving margin $\Rightarrow$ forced trade-off.

Beat 3. **Payoff.** Consistency *anti*-predicts robustness under adversarial training; η/L predicts it across datasets/threats/backbones; and the "helps" result is an FGSM artifact that dies under AutoAttack.

Pitch kit (2/3): two analogies + the killer result

Two analogies that make it click:

- **"Average colour."** A linear invariant of an image can see nothing but its *average colour* — everything about *where* things are is thrown away. That is why forcing invariance can collapse the usable margin (Ge's one-pixel example: margin drops to $2/\sqrt{d}$).
- **"Decisive and gentle."** A model is certifiably safe when it is *decisive* (big margin) *and gentle* (small Lipschitz). η/L is exactly "decisiveness per unit sensitivity" — and sensitivity must be measured in the attack's own (dual) norm.

The single result to lead with

Retrain the "invariance helps" models under their own recipe: the exactly invariant arm has *perfect* shift-consistency and looks best under FGSM (0.497 vs. 0.477), yet under AutoAttack *every* arm collapses to ≤ 0.006. The advantage was a weak-attack artifact. Under real (adversarial-training) robustness, consistency -0.88 vs. η/L $+0.88$.

Pitch kit (3/3): three questions you *will* get, with answers

Q1. **“$r_2 \geq \eta/L$ is just a loose lower bound — how can it *characterize* robustness?”**
A. As a certificate it is one-sided, yes. But (i) empirically it *predicts* robustness (correlations above), and (ii) we prove a two-sided bracket $\eta/L \leq r_2 \leq \eta/\alpha$ that pins r_2 up to a condition number $\kappa = L/\alpha$ (exact, κ=1, for linear invariants).

Q2. **“The dot / DC collapse is a toy. Does it matter on real data?”**
A. The same accounting is what the capacity-matched CIFAR / ResNet / ImageNet-100 dissections measure: forcing exact invariance collapses the margin on real data too, and η/L predicts robustness at ImageNet-100 resolution *with clean accuracy held flat*.

Q3. **“If η/L predicts robustness, just train on it — new defense?”**
A. No. It is scale-invariant, so its maximizer is the constant classifier, and Euler’s identity ties $\|\nabla M\|$ to M. Four first-order attempts all collapse the margin. It is a *diagnostic*, not a target.

ONE LINE TO REMEMBER

Don't ask whether a model is *invariant*.

Ask how much margin survives the invariance,

and how sensitive that margin is to the attack.

$$r_2 \geq \eta / L$$

Anass Al Ammiri · Technical University of Munich